Name: K R Manoj
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Software Architecture and Design Justification

Project: Alumni Mentorship & Mock Interview Platform

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Architecture Selection

We chose a **Layered Architecture** for the Alumni Mentorship & Mock Interview Platform.

Reason 1 – Separation of Concerns: The platform includes mentor matching, availability management, session booking, mock interviews, scorecards, and notifications. Separating presentation, business logic, and data access simplifies maintenance and allows individual modules to evolve without redesigning the entire system.

Reason 2 – Transactional Booking Reliability: Booking and availability management require atomic slot locking to prevent double-booking when multiple students request the same mentor slot concurrently. A shared persistence layer with transaction-safe operations supports consistent booking records and reliable availability checks.

Security Advantage: Authentication and profile management can enforce role-based access control so students access their own permitted information and mentors' access authorized sessions and scorecards. Sensitive profile and feedback data should be protected through secure communication and appropriate access controls.

Performance Benefit: Keeping application modules within one deployable application reduces unnecessary inter-service network calls. Efficient database queries and appropriate indexing can improve mentor search, availability checks, and scorecard retrieval.

Conclusion: Layered architecture provides a maintainable design for the platform while supporting its core requirements for reliable booking, controlled data access, and efficient user interactions.

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